An Empirical Analysis of Hallucination Behavior in LLMs in the Medical Domain

Aidan Perez
MS in AI, Boston University
aperez00@bu.edu

Dang Dinh
MS in Data Science, Boston University
dddinh@bu.edu

Kaveri Goel
MS in AI, Boston University
kvrgl@bu.edu

Ying Huang
MS in CS, Boston University
yinghy@bu.edu

Abstract

Large Language Models (LLMs) have demonstrated remarkable capabilities in generating fluent, contextually relevant text. However, they frequently produce hallucinations — outputs that appear plausible but are factually incorrect, unsupported, or fabricated. This problem is especially critical in high-stakes domains such as medicine, where incorrect information can have serious real-world consequences. While hallucination has been studied in general NLP settings, relatively little empirical work systematically characterizes hallucination behavior specifically within the medical domain — where ground truth is verifiable, the cost of error is high, and LLMs are increasingly being adopted (e.g., in clinical decision support and patient-facing chatbots). In this paper, we present a comprehensive empirical analysis of hallucination behavior in LLMs within the medical domain. We evaluate multiple state-of-the-art LLMs on a diverse set of medical tasks, including question answering, summarization, and clinical note generation. We systematically categorize the types of hallucinations observed, analyze their frequency and severity, and identify common patterns and triggers. Our findings provide critical insights into the nature of hallucination in medical LLMs, highlight the risks associated with their deployment in clinical settings, and suggest directions for future research aimed at mitigating hallucination and improving the reliability of LLMs in healthcare applications.

1 Introduction

Large Language Models (LLMs) have revolutionized natural language processing (NLP) by demonstrating unprecedented capabilities in generating coherent, contextually relevant text across a wide range of tasks. However, a significant challenge that has emerged with the deployment of LLMs is their propensity to produce hallucinations — outputs that may be fluent and plausible but are factually incorrect, unsupported by evidence, or entirely fabricated. This issue is particularly concerning in high-stakes domains such as medicine, where the dissemination of incorrect information can lead to serious consequences for patient care and safety. Despite the growing body of research on hallucination in general NLP contexts, there remains a paucity of systematic empirical analyses that specifically characterize hallucination behavior in medical LLMs. Given the increasing adoption of LLMs in healthcare applications, such as clinical decision support systems and patient-facing chatbots, it is imperative to understand the nature and implications of hallucination in this domain. In this paper, we present a comprehensive empirical analysis of hallucination behavior in LLMs within the medical domain. We evaluate multiple state-of-the-art LLMs across a diverse set of medical tasks, including question answering, summarization, and clinical note generation. Our analysis categorizes the types of hallucinations observed, quantifies their frequency and severity, and identifies common patterns and triggers. The insights gained from our study provide critical guidance for the responsible deployment of LLMs in healthcare settings and inform future research aimed at mitigating hallucination and enhancing the reliability of medical LLMs. Solving this problem is crucial for ensuring that LLMs can be safely and effectively integrated into clinical workflows, ultimately improving patient outcomes and advancing the field of medical AI. This is a hard problem to solve, and we hope that our work will inspire further research in this area. The reason that this problem is so hard to solve is that it requires a deep understanding of the underlying language model, as well as the ability to identify and mitigate hallucinations in a way that does not compromise the model's ability to generate useful and relevant text. Additionally, the medical domain is particularly challenging due to the complexity and nuance of medical language, as well as the high stakes involved in providing accurate information. We hope that our work will provide a foundation for future research in this area, and that it will ultimately lead to the development of more reliable and trustworthy LLMs for use in healthcare applications.

2 Related Work

Hallucination in large language models has been studied extensively in general NLP settings, with surveys categorizing hallucinations as either intrinsic (contradicting the source) or extrinsic (introducing unsupported information). This taxonomy informs our own qualitative error analysis. Prior work on faithfulness in abstractive summarization demonstrates that models frequently generate plausible but unsupported content — a pattern we expect to observe in the medical domain as well.

In the medical domain, recent work evaluating large language models on medical question answering finds that while models achieve competitive performance on benchmarks such as MedQA, they remain prone to confident errors on complex clinical reasoning tasks. Domain-specific fine-tuning reduces but does not eliminate hallucination, motivating our comparison of general-purpose models (Llama3) against medically fine-tuned variants (BioMistral, BioGPT).

Retrieval-augmented generation (RAG) has been proposed as an inference-time mitigation strategy for hallucination. By grounding model outputs in retrieved evidence, RAG reduces reliance on parametric knowledge, which is particularly beneficial in knowledge-intensive domains like medicine. Our work builds on this by directly comparing RAG against zero-shot and chain-of-thought baselines on standardized medical benchmarks.

3 Progress

3.1 Methods

To conduct our empirical analysis of hallucination behavior in medical LLMs, we have selected two widely used datasets: MedQA and PubMedQA. The MedQA dataset consists of multiple-choice questions derived from the United States Medical Licensing Examination (USMLE), covering a broad range of medical topics and requiring complex reasoning and domain-specific knowledge. The PubMedQA dataset, on the other hand, consists of yes/no questions based on abstracts from the PubMed database, which require the model to extract relevant information from the provided abstracts to generate accurate responses. We have chosen these datasets because they represent different types of medical tasks and require varying levels of reasoning and knowledge, allowing us to comprehensively evaluate the hallucination behavior of LLMs across a spectrum of medical applications. We have implemented a systematic evaluation framework that includes regular prompting, chain-of-thought prompting, and Retrieval-Augmented Generation (RAG) to assess the performance of the LLMs on these datasets. Regular prompting involves providing the model with a straightforward prompt to generate a response, while chain-of-thought prompting encourages the model to generate intermediate reasoning steps before arriving at a final answer. RAG combines retrieval-based methods with generative models, allowing the model to access external knowledge sources to generate more accurate and relevant responses. By comparing the outputs generated by the LLMs under these different prompting strategies, we aim to identify patterns of hallucination and evaluate the effectiveness of various techniques for mitigating this issue in medical LLMs.

3.2 Preliminary Findings

We conducted a preliminary comparison of Meta-Llama-3-8B-Instruct under two inference conditions on the MedQA benchmark: standard zero-shot prompting and chain-of-thought (CoT) prompting. On 500 evaluation samples, zero-shot prompting yielded 302 correct predictions (60.4% accuracy), a hallucination rate of 39.6%, 4 parse failures, and an average BERTScore F1 of 0.819. In contrast, CoT prompting yielded 282 correct predictions (56.4% accuracy), a hallucination rate of 43.6%, 18 parse failures, and an average BERTScore F1 of 0.814. These results indicate that, in our current MedQA setting, CoT does not outperform the zero-shot baseline for Llama3. Instead, it is associated with slightly lower accuracy, a higher hallucination rate, and reduced output stability.

Inspection of the raw generations suggests that this pattern may be partly explained by differences in output format between the two prompting conditions. Under zero-shot prompting, Llama3 usually provides an answer letter that can be extracted reliably, although many responses still include additional explanatory text. Under CoT prompting, the model more frequently produces longer reasoning traces and explicit answer markers, but these generations are also more likely to contain repetition, trailing phrases, and other formatting artifacts. As a result, CoT outputs are more vulnerable to parsing errors in the current pipeline.

Taken together, these findings suggest that explicit reasoning prompts do not automatically improve performance on medical multiple-choice QA. In our current setup, longer CoT generations may create additional opportunities for noisy intermediate reasoning and answer extraction failure. We therefore treat this as a preliminary result tied to the current prompting and evaluation design, rather than as a general claim that CoT is ineffective in medical QA. In future work, we plan to investigate whether stricter output constraints and more robust answer extraction can improve the reliability of CoT prompting.

3.3 RAG Preliminary Findings

We evaluated three models under Retrieval-Augmented Generation (RAG) conditions on the PubMedQA benchmark using BM25 retrieval with top-2 abstracts. On 450 evaluation samples, Meta-Llama-3-8B-Instruct achieved 73.4% accuracy with a hallucination rate of 28.9% and an average BERTScore F1 of 0.850. BioMistral-7B achieved 52.9% accuracy with a hallucination rate of 54.9% and BERTScore F1 of 0.816. BioGPT achieved 54.4% accuracy but with a hallucination rate of 89.1% and a parse failure rate of 80%, indicating the model struggled to generate structured yes/no/maybe responses.

These preliminary findings suggest that general-purpose models such as Llama3 benefit more from RAG than smaller domain-specific models. BioGPT's high parse failure rate indicates that the current prompting strategy may not be well-suited for smaller generative models, which we plan to address in future experiments. Comparison between RAG and zero-shot conditions on the same PubMedQA questions remains pending and will be included in the final report.

4 Results

4.1 MedQA: Zero-Shot vs. Chain-of-Thought

Table 1 summarises the performance of Meta-Llama-3-8B-Instruct on the MedQA benchmark under zero-shot and chain-of-thought (CoT) prompting (500 samples).

Metric	Zero-Shot	CoT
Correct	302	282
Accuracy	60.4%	56.4%
Hallucination Rate	39.6%	43.6%
Parse Failures	4	18
Avg. BERTScore F1	0.819	0.814

Table 1: MedQA results for Llama-3-8B-Instruct (500 samples).

Zero-shot prompting outperforms CoT on every metric. CoT produces longer reasoning traces that introduce more formatting artifacts, repetition, and parse failures, leading to a higher measured hallucination rate. These results suggest that explicit reasoning prompts do not automatically improve performance on medical multiple-choice QA.

4.2 PubMedQA: Retrieval-Augmented Generation

We evaluated four models under RAG conditions on PubMedQA using BM25 retrieval with top-2 abstracts. Table 2 reports the results.

Model	Acc.	Halluc.	BERT F1
Llama-3-8B	73.4%	28.9%	0.850
BioMistral-7B	52.9%	54.9%	0.816
BioGPT	54.4%	89.1%	—
Qwen2.5-3B	19.0%	81.0%	—

Table 2: PubMedQA RAG results. Llama-3 and BioMistral evaluated on 450 samples; Qwen2.5-3B-Instruct on 200 samples.

Llama-3-8B-Instruct achieves the highest accuracy and lowest hallucination rate, suggesting that larger, general-purpose instruction-tuned models benefit most from retrieval augmentation. BioGPT and Qwen2.5-3B-Instruct exhibit very high hallucination rates; inspection reveals that both models overwhelmingly predict "maybe" (Qwen2.5-3B predicted "maybe" for 175 out of 200 questions), failing to commit to definitive yes/no answers even when the retrieved evidence supports one.

4.3 Prediction Distribution Analysis

Qwen2.5-3B-Instruct shows a strong bias toward hedging. On PubMedQA, 87.5% of its predictions were "maybe," compared to a gold-label "maybe" rate of only 11.0%. On MedQA (zero-shot), the model exhibited a preference for option D (45.8%

of predictions), while CoT prompting shifted the distribution heavily toward option A (62.5%). This sensitivity of the answer distribution to the prompting strategy—without a corresponding change in accuracy—suggests that the model's predictions are driven more by surface-level prompt cues than by genuine medical reasoning.

4.4 Summary of Findings

Across all experiments, three key findings emerge:

1. **Larger instruction-tuned models hallucinate less under RAG.** Llama-3-8B-Instruct achieves 73.4% accuracy on PubMedQA with RAG, substantially outperforming both domain-specific and smaller models.

2. **CoT prompting does not reliably reduce hallucination.** On MedQA, CoT yields lower accuracy and higher hallucination rates than zero-shot prompting, largely due to increased output noise and parse failures.

3. **Smaller models exhibit systematic prediction biases.** Qwen2.5-3B-Instruct and BioGPT show degenerate prediction distributions (e.g., overwhelming "maybe" bias), indicating that these models lack the capacity to leverage retrieved evidence effectively for medical QA.

4.5 Future Work

Our future work will focus on the following directions:

- **Improve the robustness of CoT prompting.** Our current results suggest that chain-of-thought (CoT) prompting is more vulnerable to formatting noise, repetition, trailing text, and parse failures. To address this issue, we plan to test stricter output constraints, shorter reasoning prompts, and alternative prompt templates that require the model to end with one clearly formatted final answer. We also plan to improve the answer extraction rules so that small variations in output style do not overly affect the measured hallucination rate.

- **Continue fine-tuning experiments.** We will continue the fine-tuning component of the project and investigate whether combining fine-tuning with retrieval can improve reliability beyond either approach alone. In particular, we want to study whether domain-specific fine-tuning can reduce hallucination further when paired with RAG.

- **Evaluate additional models.** We plan to test newer and more strongly instruction-tuned models, as well as additional medically adapted models, to examine whether the current CoT limitations are specific to the models we have already tested.

- **Fix ground-truth extraction for MedQA on Qwen2.5-3B.** Our current MedQA evaluation pipeline for Qwen2.5-3B-Instruct produced incorrect ground-truth labels due to an answer-matching issue in the dataset loader. We plan to resolve this and re-run the evaluation to obtain valid accuracy figures for this model on MedQA.